

Homework 2. Baseline Models

You can get up to 10 points for this homework.

Question 1. Table 1 on the next page contains data on the number of airline miles flown in the United Kingdom. For the following tasks use Python and submit one .ipynb file incorporating all your answers.

- ☐ A Create a .csv file out of the given data and put it online (you will use a link to it in your program).
- ☐ B Create a time series plot to visualise the data.
- ☐ C Decompose the time series into three components (trend, seasonal and residuals). Present the decomposition together with the observed data in the same figure. What conclusions can you make about the time series by looking at the graphs?
- ☐ D Divide the data into a train set and a test set.
- ☐ E Implement the four baseline models: historical mean, last year's mean, last known value and naive seasonal forecast.
- ☐ F Compare these baselines models. Which of them looks more efficient for our time series?

TABLE 1. United Kingdom Airline Miles Flown

Month	Miles (In Millions)	Month	Miles (In Millions)
Jan-1964	7.269	Jul-1967	12.222
Feb-1964	6.775	Aug-1967	12.246
Mar-1964	7.819	Sep-1967	13.281
Apr-1964	8.371	Oct-1967	10.366
May-1964	9.069	Nov-1967	8.730
Jun-1964	10.248	Dec-1967	9.614
Jul-1964	11.030	Jan-1968	8.639
Aug-1964	10.882	Feb-1968	8.772
Sep-1964	10.333	Mar-1968	10.894
Oct-1964	9.109	Apr-1968	10.455
Nov-1964	7.685	May-1968	11.179
Dec-1964	7.682	Jun-1968	10.588
Jan-1965	8.350	Jul-1968	10.794
Feb-1965	7.829	Aug-1968	12.770
Mar-1965	8.829	Sep-1968	13.812
Apr-1965	9.948	Oct-1968	10.857
May-1965	10.638	Nov-1968	9.290
Jun-1965	11.253	Dec-1968	10.925
Jul-1965	11.424	Jan-1969	9.491
Aug-1965	11.391	Feb-1969	8.919
Sep-1965	10.665	Mar-1969	11.607
Oct-1965	9.396	Apr-1969	8.852
Nov-1965	7.775	May-1969	12.537
Dec-1965	7.933	Jun-1969	14.759
Jan-1966	8.186	Jul-1969	13.667
Feb-1966	7.444	Aug-1969	13.731
Mar-1966	8.484	Sep-1969	15.110
Apr-1966	9.864	Oct-1969	12.185
May-1966	10.252	Nov-1969	10.645
Jun-1966	12.282	Dec-1969	12.161
Jul-1966	11.637	Jan-1970	10.840
Aug-1966	11.577	Feb-1970	10.436
Sep-1966	12.417	Mar-1970	13.589
Oct-1966	9.637	Apr-1970	13.402
Nov-1966	8.094	May-1970	13.103
Dec-1966	9.280	Jun-1970	14.933
Jan-1967	8.334	Jul-1970	14.147
Feb-1967	7.899	Aug-1970	14.057
Mar-1967	9.994	Sep-1970	16.234
Apr-1967	10.078	Oct-1970	12.389
May-1967	10.801	Nov-1970	11.594
Jun-1967	12.953	Dec-1970	12.772